

# Student Engagement Classification based on Physiological, Behavioral and Environmental Factors

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## 1 Introduction

This project aims to analyse data on student physiological and behavioral metrics alongside environmental factors to see if their engagement level in class can be determined. Student engagement is vital for the learning process [1]. Engaged students, 'actively explore ideas confidently with others,' 'feel confident contributing ideas,' 'ask questions in class,' among other benefits [2]. A deeper understanding of the factors that affect this could provide insight into improving education.

## 2 Dataset

The emotional monitoring dataset [3] contains recorded data of a group of student's physiological and behavioral responses to the educational environments in order to track their engagement levels.

There are also environmental features present; the AmbientNoiseLevel and LightingLevel. They quantify the amount of noise in decibels (dB) and light intensity (lux) present in the environment respectively.

Features for student physiological and behavioral responses include HeartRate (beats per minute, bpm), SkinConductance (microsiemens,  $\mu S$ ), Electroencephalography (EEG), Temperature (degrees, C), PupilDiameter (millimeters, mm), SmileIntensity (0.0 - 1.0), FrownIntensity (0.0 - 1.0), CortisolLevel (micrograms per deciliter,  $\mu g/dL$ ), ActivityLevel (steps).

The target column is the EngagementLevel which categorises student engagement into Highly Engaged, Moderately Engaged and Disengaged levels.

### 2.1 Pre-processing

To prepare the dataset suitable for the research, several preprocessing steps were taken. Initially the dataset was cleaned by removing duplicate rows and any missing values (NaNs). To streamline the classification process, the 'EngagementLevel' feature, which originally contained three categories (1: Highly Engaged, 2: Moderately Engaged, 3: Disengaged), was simplified into two categories. An 'Engaged' category combining 'Highly Engaged' and 'Moderately Engaged', and a 'Disengaged' category which solely contained the original 'Disengaged' category. The decision to combine moderate and high engagement levels was taken to give more leeway towards general engagement over a high standard of engagement which may be difficult to assess.

Furthermore, the 'EmotionalState' and 'CognitiveState' columns were dropped as they were not recorded in the data card documentation on Kaggle and deemed to likely be subjective analyses of each student, potentially adding noise or bias to the data and subsequently affecting the model training and analysis.

A stratified train-test split of 80/20 was used so the original dataset proportions were similar to the proportions in the target classes.

### 2.2 Logistic Regression

A logistic regression model was implemented to evaluate performance using 5-fold cross-validation. Logistic regression is a linear classification algorithm that categorises outcomes. The model was initialised with a maximum iteration limit (max\_iter=1000) to ensure convergence and a fixed random state (random\_state=42) for reproducibility.

Model performance was assessed by dividing the dataset into five-folds for cross-validation. One fold was used for validation in each iteration, the remaining were used for training until all data points were used for both purposes.

## 2.3 Interpretation of Results

This approach produced accuracy scores of [0.59, 0.575, 0.56, 0.565, 0.645], with a mean accuracy of 58.7% and a standard deviation of 0.031. The low standard deviation indicates consistent performance, but the mean accuracy suggests the logistic regression model struggled to capture patterns effectively.

To enhance results, Principal Component Analysis (PCA) was applied to reduce dimensionality while retaining 95% of the variance. Features were standardized using StandardScaler before PCA. The first principal component captured 65.6% of the variance, with the total variance explained reaching 96.1%, reducing the dataset to 6 components while retaining most of the information.

A logistic regression model trained on the PCA-transformed data achieved cross-validation scores of [0.635, 0.585, 0.61, 0.64, 0.645], with a mean accuracy of 62.3%. PCA improved the model's performance and maintained stable results across folds, though further refinement is needed to achieve higher accuracy.

## 2.4 Random Forest Classifier

The model was changed to a Random Forest Classifier using 5-fold cross-validation. Random Forest Classifiers create decision trees during training and aggregate their predictions.

To assess performance, the dataset was split into five folds for cross-validation. In each iteration, one fold served as the validation set, and the other four were used for training. This approach yielded the following accuracy scores: [0.865, 0.885, 0.865, 0.825, 0.905]. The mean accuracy was 86.9%, and the standard deviation was 0.027, reflecting consistent performance across folds.

## 2.5 Interpretation of Results

The Random Forest Classifier demonstrated strong performance. The low standard deviation indicates that the model performed consistently regardless of the specific data splits. This suggests the classifier effectively captured the underlying patterns in the data while maintaining stability.

The hyperparameters struck a balance between bias and variance. Tree depth was limited and a minimum number of samples being required for splits and leaves avoided overfitting. Further tuning may offer additional improvements.

# 3 Conclusion

This study explored the use of physiological, behavioral, and environmental data to classify student engagement. Logistic Regression, as a baseline linear model, achieved 58.7% accuracy, highlighting its limitations with non-linear data.

Switching to a Random Forest Classifier improved accuracy significantly. The mean was accuracy across data splits was raised to 86.9%, suggesting the features in the datasets have complex non-linear relationships and interactions with each other to influence student engagement. Simplifying the target variable into two categories—Engaged and Disengaged—focused the classification task but lost distinctions between moderate and high engagement. Dropping undocumented features like "EmotionalState" and "CognitiveState" likely reduced noise, improving model reliability.

The model identified strong predictors of engagement, such as heart rate and ambient noise levels. These insights could have practical applications, such as designing quieter learning environments or using physiological data to monitor student engagement in real time. For example, real-time monitoring could help educators detect disengagement early and adjust teaching strategies accordingly. Of course, this may have some ethical implications but that is out of scope for the project.

Future research could refine models, carefully incorporate subjective features, and test these approaches in real classrooms to enhance both accuracy and practical impact.

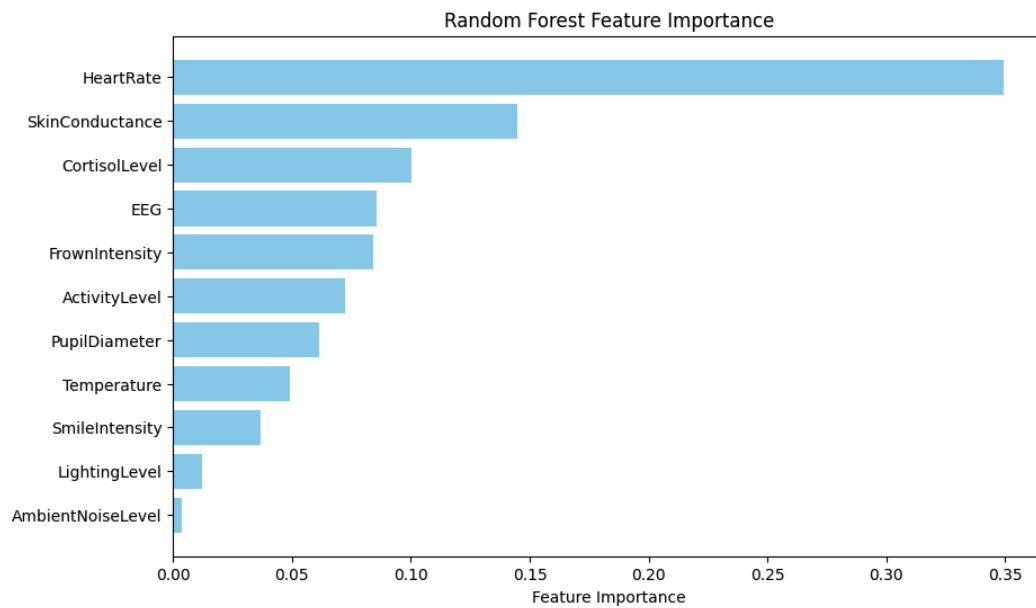


Figure 1: The figure shows the relative importance of each feature in determining the classification from the features. HeartRate was the most important at 0.35. Environmental features were the lowest. This may be due to the more indirect effect they have on measuring student emotional stability compared to physiological features which are a more direct indicator of stress or lack of focus.

## References

- [1] Jim Parsons and Leah Taylor. Improving student engagement. *Current issues in education*, 14(1), 2011.
- [2] V Trowler. Student engagement literature review. *Higher Education Academy*, 2010.
- [3] Ziya. Emotional monitoring dataset. Kaggle, Nov 2024. <https://www.kaggle.com/datasets/ziya07/emotional-monitoring-dataset>.

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